

Phase. Phase in sinusoidal functions or in periodic waves has two different but closely related meanings. One is the initial angle of a sinusoidal function at its origin, and is sometimes called phase offset or phase difference. Another usage is the fraction of the wave cycle that has elapsed relative to the origin. Refer Fig. 3, where the phase difference between the red and blue waves is ' θ '.

Phase shifts are common in electronics. The amplitude, frequency and phase characteristics of a waveform are the building blocks that a signal generator uses to optimise waveforms for almost any application. These are implemented as controlled variables in all the signal generators.

Rise and fall times. Rise and fall times, also known as edge transition times, are characteristics usually described for pulses and square waves. These are the measure of the time it takes for a signal edge to make a transition from one state to another. With the advancement in technology, these values have typically gone down to nanoseconds range.

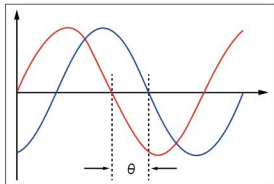


Fig. 3: Phase difference between the red and blue waves is ' θ '

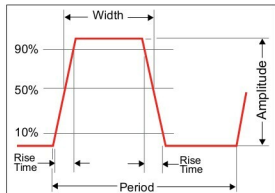


Fig. 4: A pulse and some of the characteristics associated with it

Both rise and fall times are measured between 10% and 90% points of the static voltage levels before and after the transition as shown in Fig. 4. You will need a functionality in the signal generator where you can vary the rise time and fall time independently for a waveform while testing applications such as an amplifier with asymmetrical slew rates. So look for a signal generator that supports this function.

Pulse width. Pulse width is the time elapsed between the leading and trailing edges of a pulse. Pulse width measurement is expressed as the time between the 50 per cent amplitude points of the respective edges. This is an important variable function for a signal generator, which you will need while experimenting with servo motor, LED intensity, etc. Fig. 5 shows a signal with different duty cycles.

DC offset. Many signals have their amplitude variations centred above or below 0V reference. 'Offset' voltage is the voltage between circuit ground and the centre of the signal's amplitude variation. In effect, the offset voltage expresses the DC component of a signal containing both AC and DC values. This DC component needs to be handled carefully in most of the designs that require a stimulus system like signal generator in the design phase, which can vary this component in a controlled manner to test the implementation.

Differential signals. These are two complementary signals that have equal and opposite polarity with respect to ground. Fig. 6 shows the single-ended and differential signals. In differential signal, as the signal's cycle proceeds one path becomes more positive, the other becomes more negative to the same degree. The differential architecture is good at rejecting crosstalk and noise and passing only the valid signal. Single-ended operation is a more common architecture, in which there is only one path plus ground. To design and test the systems that communicate via differential system, you need a signal generator that has this functionality.

Complex waveforms

The above-mentioned are only basic characteristics of the waveform and corresponding specifications of the signal generator. In operational electronic systems, waveforms rarely look like the examples explained above. Certain clock and carrier signals are pure, but most other waveforms will exhibit some unintended distortion. Some waveforms may even include elements of sine, squares, steps and pulses. To test, debug and design such real applications, more sophisticated stimulus signals are required than just simple waveforms. Do check whether the signal generator has these features because you are really going to need them.

Modulated signals. The most commonly required complex waveform is a modulated signal where the modulation parameters can be controlled. In

modulated signals, amplitude, phase and/or frequency variations embed lower-frequency information into a carrier signal of higher frequency. The resulting signals may convey anything from speech to data to video. Such waveforms can be a challenge to reproduce, unless the signal generator is specifically designed for it.

Frequency sweep. If you want to check the frequency response of your device, you need

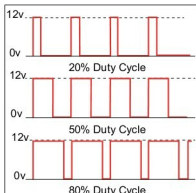


Fig. 5: A signal with different duty cycles